

ARMORED LIKHO STILL TOOLKIT SERVICE INSTALLATION

Campaign hunt for source-observed Windows services created by Telegram-theft and audio-surveillance implants.

HUNTING QUERY

CONFIDENCE: MEDIUM

REPORT DATE

14 August 2026

VERSION

v1.0

PRIMARY TELEMETRY

Microsoft Sentinel ·
SecurityEvent · 4697

ATT&CK

T1543.003

Evidence-led dossier. Query status remains an untested implementation sketch until executed in the target environment.

INDEPENDENT RESEARCH FORMAT · CONTROLLED UI/UX MASTER

SOURCE-CONFIRMED BEHAVIOR

retain facts, isolate interpretation

SecurityEvent · 4697

direct telemetry boundary

HUNT · VALIDATE · TUNE

no automatic malicious verdict



EXECUTIVE ASSESSMENT

Decision-grade summary for detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Retain the two source-observed service names as a campaign hunt.

Kaspersky observed Still Sync creating the TReLoad service and Still Audio creating auxhost when --console is absent. Windows Security Event 4697 directly records new-service installation and the installed service name.

OPERATIONAL VALUE

Direct persistence metadata, constrained by artifact mutability.

Event 4697 captures service creation before later execution may be obscured. The names provide immediate scoping value, but the source does not publish the exact service command and either name can change; the logic therefore remains a Medium-confidence campaign hunt.

8.6

01

OPERATIONAL VALUE



Medium

02

DETECTION CONFIDENCE



Hunting

03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACTS

Observed behavior

- The campaign was discovered in May 2026 and published 13 August 2026.
- Still Sync creates TReLoad when --console is absent.
- Still Audio creates auxhost when --console is absent.

ANALYTICAL ASSESSMENT

What the evidence supports

- Event 4697 is a direct record of new-service installation.
- The names are useful campaign pivots but are not durable behavior.
- A match is a lead, not proof of implant execution or attribution.

DETECTION ASSUMPTIONS

What must be validated

- Audit Security System Extension is enabled and Event 4697 reaches Sentinel.
- SecurityEvent populates ServiceName for Event 4697.
- Any name collision can be resolved through command, file and ownership review.

Confidence boundary

The source does not publish the exact ImagePath/service command. The query cannot prove that the service started, Still Sync stole Telegram data, Still Audio captured audio, C2 succeeded, or Armored Likho operated the host.



THREAT MECHANICS

Source-observed chain and explicit observability boundaries.

ATTACK FLOW

01

Deliver lure

A fake donation application deceives the user.

02

Decrypt payload

The Tauri dropper launches the next stage.

03

Select mode

Absence of --console enables service persistence.

04

Install service

Still Sync creates TReload; Still Audio creates auxhost.

05

Operate implant

The modules collect Telegram data or microphone audio.

Observable core

A Windows Security Event 4697 whose installed ServiceName is TReload or auxhost.

Non-observable boundary

Lure delivery, decrypted payload content, service start, Telegram session theft, microphone capture, C2 success and actor attribution.

DETECTION HYPOTHESIS

Still Toolkit persistence should leave a new-service installation record.

If the attacker repeats the documented background-mode installation, SecurityEvent should record Event ID 4697 with ServiceName TReload or auxhost. Legitimate collisions can be separated through ServiceFileName, file signer and hash, installation identity, host role, change records and neighboring process or network evidence.

REQUIRED TELEMETRY

SecurityEvent · 4697

- Windows Security Event 4697.
- Windows Security Events via AMA.
- ServiceName, host and raw-event identity.

OPTIONAL ENRICHMENT

Context that improves disposition

- ServiceFileName and service account.
- File hash, signer and origin.
- Process, service-start and network telemetry.

UNAVAILABLE HERE

Do not infer

- Exact service command from the report.
- Proof that the installed service executed.
- Telegram theft, audio capture or C2 outcome.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and quality gates for a defensible hunt.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Sentinel · SecurityEvent · 4697

SecurityEvent documents EventID, ServiceName, ServiceFileName, service metadata, subject identity, host and EventRecordId. Microsoft documents Event 4697 as generated when a new service is installed.

EVIDENCE BOUNDARY

A matching event is a lead, not a verdict.

The event establishes service-installation metadata only. The source does not justify filtering on a path, executable name, account, start type or command line.

REQUIRED AND ENRICHMENT FIELD MODEL

CATEGORY	FIELD	PURPOSE	REQUIREMENT
Service event	SecurityEvent / EventID	4697 installation event.	REQUIRED
Service identity	ServiceName	TReload or auxhost.	REQUIRED
Host	Computer, _ResourceId	Asset pivot.	REQUIRED
Event identity	TimeGenerated, EventRecordId	Raw record pivot.	REQUIRED
Service command	ServiceFileName	Triage; not filtered.	OPTIONAL
Subject	SubjectUserSid, SubjectUserName, SubjectDomainName	Installer identity.	OPTIONAL
Logon context	SubjectLogonId	Neighboring activity pivot.	OPTIONAL
Service metadata	ServiceType, ServiceStartType, ServiceAccount	Triage enrichment.	OPTIONAL

Data quality requirements

- Confirm Event 4697 collection on relevant Windows assets.
- Verify ServiceName population using a controlled install.
- Inspect complete ServiceFileName and installer identity for every match.
- Measure historical collisions without a rarity threshold.

Known failure conditions

- Audit Security System Extension is disabled.
- The connector drops Event 4697 or ServiceName.
- The actor renames the service or avoids service persistence.
- The service already exists and is modified rather than newly installed.



DETECTION LOGIC

Primary-platform logic with every condition traceable to evidence.

HUNTING QUERY

DETECTION OBJECTIVE

Find source-observed Still Toolkit service installation.

Select new-service events only when Event 4697 records one of the two service names directly documented by Kaspersky.

ANALYTICAL LOGIC

Direct filters and investigable output.

Two direct predicates: Event ID and service-name membership. No path, process, hash, domain, threshold, join, sequence, aggregation, rarity, regex, external lookup or attribution.

KQL · UNTESTED HUNT

```
// Armored Likho / Still Toolkit Windows service installation
// Classification: Hunting query - Untested implementation sketch
// Confidence: Medium
// Primary platform: Microsoft Sentinel / Log Analytics
// Required connector: Windows Security Events via AMA with Event ID 4697
// Source evidence: Kaspersky observed Still Sync creating the TReload service
// and Still Audio creating the auxhost service when --console is absent.
// The service names are mutable campaign artifacts; a match requires triage.

SecurityEvent
| where EventID == 4697
| where ServiceName in~ ("TReload", "auxhost")
| project
    TimeGenerated,
    Computer,
    EventID,
    EventRecordId,
    ServiceName,
    ServiceFileName,
    ServiceType,
    ServiceStartType,
    ServiceAccount,
    SubjectUserSid,
    SubjectUserName,
    SubjectDomainName,
    SubjectLogonId,
    Activity,
    EventSourceName,
    _ResourceId
| order by TimeGenerated desc
```

STATUS	THRESHOLD	JOIN / SEQUENCE	PRIMARY KEY	DISPOSITION
Untested hunt	None	None	Computer + EventRecordId + TimeGenerated	Manual triage

What a match means

The host recorded installation of a source-observed Still Toolkit service name and requires immediate service, file and installer review.

What a match does not prove

Malicious ImagePath, service start, implant execution, Telegram access, microphone use, C2 traffic, successful persistence or Armored Likho attribution.



TRIAGE & INVESTIGATION

Evidence collection before escalation.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the raw event.
Verify service name, complete ServiceFileName, host, subject and record ID.
- 02

Validate ownership.
Check CMDB, software inventory, package owner and change record.
- 03

Collect the service binary.
Resolve ImagePath, acquire the file and calculate hashes and signer state.
- 04

Rebuild execution.
Review service start, parent process, file creation and nearby commands.
- 05

Scope follow-on activity.
Search for tdata access, microphone use and reported network endpoints.

Escalation gate

Escalate when the service is unowned or its binary is unsigned/unknown, or when the event is corroborated by suspicious installer execution, Telegram tdata access, microphone activity, gRPC traffic or reported campaign infrastructure.

Primary benign overlap

TReload or auxhost may collide with internal software, test harnesses or unrelated packages. Tune only with verified publisher, binary hash/provenance, approved owner and stable path; never suppress a name globally without evidence.

INVESTIGATION PIVOTS

Host
Owner, role, exposure and EDR coverage.

Service
Current ImagePath, start type and account.

File
Hash, signer, origin and prevalence.

Installer
Subject logon and initiating process.

Telegram
tdata file access and application state.

Network
gRPC, reported C2 and GitHub fallback retrieval.

DISPOSITION CRITERIA

OUTCOME	MINIMUM EVIDENCE	ACTION
BENIGN	Approved, owned and repeatable activity.	Document and tune narrowly.
SUSPICIOUS	Unknown activity without corroborated malicious follow-on.	Deepen endpoint review.
LIKELY MALICIOUS	Unauthorized match plus coherent file, process, network or control-plane evidence.	Escalate incident response.



FALSE POSITIVES & VALIDATION

Operational controls required before promotion.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	WHY IT OVERLAPS	TUNING ACTION
Internal application	A package may reuse one generic service name.	Require verified owner, signer and stable ImagePath.
Security research replay	A lab may reproduce the published names.	Require approved analyst, isolated asset and test window.
Packaging or QA test	Deployment tooling may create disposable services.	Validate package identity and change record.

Baseline plan

- Search the full available SecurityEvent retention.
- Inspect every matching raw event and resolved ImagePath.
- Measure field completeness and connector coverage.
- Do not add a count or rarity threshold.

Validation procedure

- Enable and confirm Audit Security System Extension in an isolated lab.
- Create and remove disposable services using both names.
- Verify the unrelated-name negative case.
- Review historical matches with platform and software owners.

TEST CASES

CASE	EXPECTED RESULT
Install disposable service named TReload	Expected Event 4697 match with populated fields.
Install disposable service named auxhost	Expected Event 4697 match with populated fields.
Install unrelated service name	No match.
Process named TReload without service installation	No match; validates event boundary.

Hunting Initial query	Candidate Current status	Production Measured precision	Optimized Stable feedback
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Success

Both controlled service installs match with complete event identity, the negative case does not, and historical collisions are absent or explainable.

Rejection

ServiceName is missing or legitimate software frequently uses either name without a durable owner/path discriminator.

Failure

A controlled service install produces no usable SecurityEvent record.



ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

MITRE ATT&CK MAPPING

TECHNIQUE	NAME	COVERAGE STATEMENT
T1543.003	Create or Modify System Process: Windows Service	Directly covers creation of the TReload and auxhost services used by Still Sync and Still Audio to remain active in the background.

Detection coverage summary

Strong for the exact source-observed service names at creation time; weak for renamed services, existing-service modification, console mode, Registry-only telemetry or missing 4697 audit coverage.

Residual risk

The actor can rename the services, run with --console, modify an existing service or use another persistence method. Collection and C2 behaviors require separate telemetry.

ANALYST CONCLUSION

A useful campaign hunt that deliberately stops short of a durable detection.

Kaspersky directly documents the two service names and Microsoft documents Event 4697 plus the required Sentinel fields. That supports a Medium-confidence Hunting query. The absence of an exact service command and the mutability of names prevent Production classification.

REFERENCES

- Kaspersky GREAT, "Armored Likho expands its cyber-espionage toolkit," 13 August 2026; campaign discovered May 2026. <https://securelist.com/armored-likho-still-toolkit/121033/>
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- MITRE ATT&CK, T1543.003. <https://attack.mitre.org/techniques/T1543/003/>

DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
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Publication control

Eight A4 pages. URLs are defanged. No customer data, credentials, HTML source or rendering assets are included in the public repository.